

AI / NLP PRACTICAL

Explore the Hugging Face Library and Load a Pre-trained Transformer Model

Aim

To explore the Hugging Face Transformers library and load a pre-trained Transformer model for performing sentiment analysis on text data.

Requirements

- Google Colab
- Python
- Hugging Face Transformers library
- PyTorch
- Internet connection

Theory

Hugging Face is an open-source platform that provides tools, datasets, and pre-trained models for Artificial Intelligence and Natural Language Processing (NLP). Its Transformers library provides access to powerful models such as BERT, DistilBERT, RoBERTa, GPT, and T5.

Pre-trained Transformer models can be used for tasks such as sentiment analysis, text classification, translation, text generation, question answering, and summarization. In this experiment, the `pipeline()` function is used to load a pre-trained sentiment analysis model without training a model from scratch.

Algorithm

1. Open Google Colab and create a new notebook.
2. Install the Transformers and PyTorch libraries.
3. Import the pipeline function from the Transformers library.
4. Create a sentiment-analysis pipeline.
5. Enter a sample text as input.
6. Pass the input text to the pre-trained model.
7. Display the predicted sentiment and confidence score.
8. Verify the result.

Procedure

1. Open Google Colab in a web browser.
2. Run the installation command in a code cell.
3. Import pipeline from transformers.
4. Load the pre-trained sentiment analysis model.
5. Provide a sentence as input.
6. Run the model and observe the predicted label and score.

Program

```
# Install required libraries
!pip install -q transformers torch

# Import pipeline
from transformers import pipeline

# Load a pre-trained sentiment analysis model
classifier = pipeline("sentiment-analysis")

# Input text
text = "I love Artificial Intelligence and Machine Learning."

# Perform sentiment analysis
result = classifier(text)

# Display result
print("Input:", text)
print("Output:", result)
```

Sample Input

I love Artificial Intelligence and Machine Learning.

Sample Output

Input: I love Artificial Intelligence and Machine Learning.
Output: [{'label': 'POSITIVE', 'score': 0.99}]

Note: The exact confidence score may vary depending on the model version and runtime.

Explanation of the Program

- **!pip install -q transformers torch** – Installs the Hugging Face Transformers library and PyTorch.
- **from transformers import pipeline** – Imports the pipeline function.
- **pipeline('sentiment-analysis')** – Loads a pre-trained sentiment analysis model.
- **text** – Stores the sentence that will be analyzed.
- **classifier(text)** – Sends the text to the model and obtains the prediction.
- **label** – Shows whether the sentiment is POSITIVE or NEGATIVE.
- **score** – Shows the confidence of the model's prediction.

Result

The Hugging Face Transformers library was successfully installed and explored. A pre-trained Transformer model was successfully loaded and used to perform sentiment analysis on the given text.

Conclusion

Hugging Face provides an easy way to use powerful pre-trained Transformer models for NLP applications. Using the pipeline() function, sentiment analysis can be performed with only a few lines of Python code, without training a model from scratch.